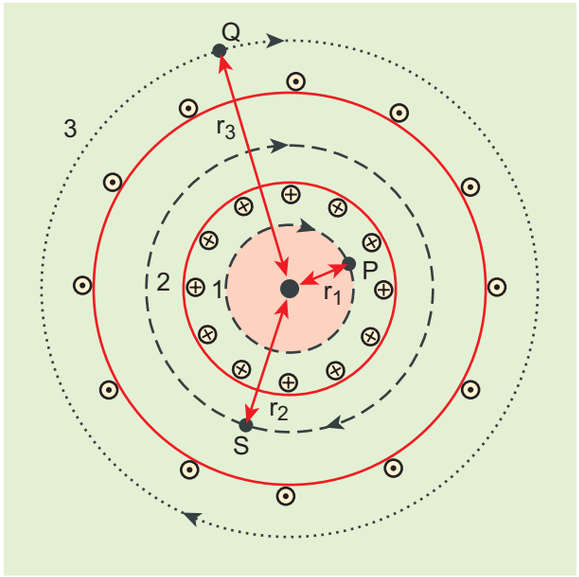


Figure 3.43 Toroid – Amperian loop

Figure 3.43. For simplicity, we take circular loop so that the length of the loop is its circumference.

$$L_1 = 2\pi r_1$$

Ampère's circuital law for the loop 1 is

$$\oint_{loop1} \vec{B}_P . \vec{dl} = \mu_0 I_{enclosed}$$

Since the loop 1 encloses no current, $I_{enclosed} = 0$

$$\oint_{loop1} \vec{B}_P . \vec{dl} = 0$$

This is possible only if the magnetic field at point P vanishes i.e.

$$\vec{B}_P = 0$$

(b) Open space exterior to the toroid

Let us calculate the magnetic field B_Q at point Q. We construct an Amperian loop 3 of radius r_3 around the point Q as shown in Figure 3.43. The length of the loop is

$$L_3 = 2\pi r_3$$

Ampère's circuital law for the loop 3 is

$$\oint_{loop3} \vec{B}_Q . \vec{dl} = \mu_0 I_{enclosed}$$

Since in each turn of the toroid loop, current coming out of the plane of paper is cancelled by the current going into the plane of paper. Thus, $I_{enclosed} = 0$

$$\oint_{loop3} \vec{B}_Q . \vec{dl} = 0$$

This is possible only if the magnetic field at point Q vanishes i.e.

$$\vec{B}_Q = 0$$

(c) Inside the toroid

Let us calculate the magnetic field B_S at point S by constructing an Amperian loop 2 of radius r_2 around the point S as shown in Figure 3.43. The length of the loop is

$$L_2 = 2\pi r_2$$

Ampere's circuital law for the loop 2 is

$$\oint_{loop2} \vec{B}_S . \vec{dl} = \mu_0 I_{enclosed}$$

Let I be the current passing through the toroid and N be the number of turns of the toroid, then

$$I_{enclosed} = NI$$

Therefore,

$$\oint_{loop2} \vec{B}_S . \vec{dl} = \oint_{loop2} B_S \, dl \cos\theta = B_S 2\pi r_2$$

$$\oint_{loop2} \vec{B}_S . \vec{dl} = \mu_0 NI$$

$$B_S = \mu_0 \frac{NI}{2\pi r_2}$$

If the number of turns per unit length $n = \frac{N}{2\pi r_2}$, then the magnetic field at point S is

$$B_S = \mu_0 nI \qquad (3.53)$$

12th Physics_Vol-1_Unit 3_English Medium.indd 167 23-12-2022 15:03:00